PHYSICAL CHEMISTRY

GUIDED REVISION PLAN-SCORE ADVANCED

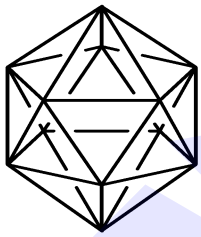
Topic : Thermodynamics & Thermochemistry (Question Bank # 03)

MM : 67 **Time : 45 Min.**

SECTION–I : (i) Only One option correct Type

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **3(–1)**

1. Boron exist in different allotropic forms. All allotropic form contains icosahedral units (icosahedral is a regular shape with 12 corners & 20 faces) with boron atoms at all 12 corners and all bonds are equivalent.

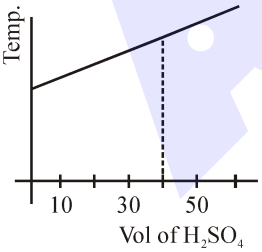


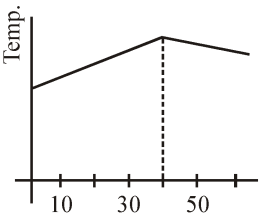
Icosahedral

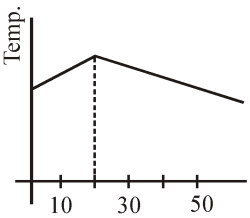
Calculate $|\Delta H|$ (in kJ) per mole of boron atoms forming gaseous icosahedral if ΔH_{BE} (B –B)= 200 kJ/mol.

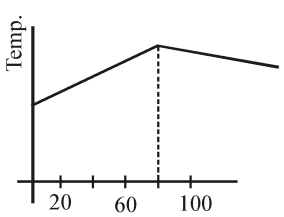
(A) 1000 (B) 600 (C) 500 (D) 250

2. In an experiment the enthalpy of neutralisation of sodium hydroxide with sulphuric acid, 50 cm^3 of 0.4M sodium hydroxide were titrated thermometrically with 0.25M sulphuric acid. Which of the following plots give the most probable correct representation temperature of solution (initial temperature of NaOH & H_2SO_4 are same).

(A) 

(B) 

(C) 

(D) 

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3. 10 litre of an ideal gas at 25 atm & 27ºC is expanded isothermally to 1atm against a constant external pressure of 760 torr. Calculate work done by the gas in litre-atm

(A) 240 (B) 250 (C) 800 (D) 400

4. Which of the following reactions is spontaneous only at relatively low temperature -

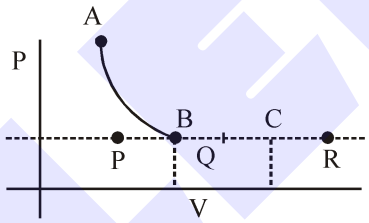
(A) $NH_4Br(s) + 188\ kJ \rightarrow NH_3(g) + Br_2(l)$

(B) $NH_3(g) + HCl(g) \rightarrow NH_4Cl(s) + 176kJ$

(C) $2H_2O_2(l) \rightarrow 2H_2O(l) + O_2(g) + 196\ kJ$

(D) Both (B) & (C)

5. For an ideal gas, three adiabatic processes are carried out upto same final pressure from same initial state. If adiabatic reversible process ends up at point 'B' and adiabatic single step irrversible process ends up at point 'C', then adibatic free expansion upto same final pressure will end up at -



(A) A (B) P

(C) Q (D) R

6. Which of the options contain correct set of parameters which are both amount and path independent parameters.

(I) Internal energy

(II) Specific enthalpy

(III) Heat exchange at constant pressure condition

(IV) Molar heat exchange at constant volume condition

(V) Temperature

(VI) Work involved

(A) Only II, IV & V (B) All option except VI

(C) All options except III, IV & VI (D) Only II & V

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(ii) One or more options correct Type

This section contains **5 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. **4(–1)**

7. Assume ideal gas behaviour for all the gases considered & vibrational degrees of freedom to be active. Separate equimolar samples of Ne, O_2, SO_2 & CH_4 were subjected to a two step process as mentioned. Initially all are at same state of temperature & pressure.

Step-I ⟶ All undergo reversible adiabatic expansion to attain same final volume, which is double the original volume thereby causing the decreases in their temperature.

Step-II ⟶ After step I all are given appropriate amount of heat isochorically to restore the original temperature.

Mark the correct option(s)

(A) Due to step I only, the decrease in temperature will be maximum for Ne.

(B) During step II, heat given will be minimum for CH_4

(C) There will be no net change in internal energy for any of the gas after both the steps of process are completed.

(D) The P–V graph of CH_4 & SO_2 will be same for overall process.

8. Choose the correct statement(s)

(A) During adiabatic expansion of an ideal gas magnitude of work obtained is equal to ΔH of gas.

(B) For same change in temperature of ideal gas through adiabatic process, magnitude of W will be same in reversible as well as irreversible process

(C) During boiling of H_2O at 1 atm, 100ºC kinetic energy of H_2O molecules increases.

(D) For same change in volume of ideal gas, ΔU is more in adiabatic expansion than in isobaric expansion if expansion starts with same intial state.

9. Select the correct options(s):

(A) $\Delta H_f^\circ [H_2O(l)]$ is zero at 25°C

(B) $\Delta S_f^\circ [O_2(g)]$ is zero at 25°C

(C) $S^\circ[O_2(g)]$ is zero at 25°C

(D) Entropies of aqueous ions can be negative or positive

10. Salect the ***incorrect*** statement (s) -

(A) $\Delta H_{atomisation}$ of graphite is equal to ΔH_f [C(g)]

(B) ΔH_{comb} of 'H' atom is equal to ΔH_f $[H_2O(l)]$

(C) ΔH_f $[H_2O(l)]$ is zero

(D) ΔH_{comb} of graphite is equal ΔH_f [CO(g)]

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11. Which of the following is correct option(s)-

($\in$: Bond enthalpy, $\Delta_a H$: enthalpy of atomisation)

(A) $(\Delta_f H)_{H(g)} = \frac{1}{2}(\Delta_a H)_{H_2(g)} = \frac{1}{2} \in_{H-H}$

(B) $(\Delta_f H)_{C(g)} = (\Delta_a H)_{C(gr,s)} = (\Delta_{sub} H)_{C(gr,s)}$

(C) $(\Delta_f H)_{Br(g)} = \frac{1}{2}(\Delta_a H)_{Br_2(l)} = \frac{1}{2} \in_{Br-Br}$

(D) $(\Delta_f H)_{I(g)} = (\Delta_a H)_{I_2(s)} = (\Delta_{sub} H)_{I_2(s)}$

(iii) Paragraph Type

This section contains **1 paragraphs** each describing theory, experiment, data etc. **3 questions** relate to one paragraph with these questions. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D). **3(0)**

Paragraph for Q. No. 12 to 14

Study the following thermochemical equation :

(a) $N_2O(g) + 3H_2(g) \rightarrow H_2O(l) + N_2H_4(l)$; $\Delta H = -76$ kcal

(b) $4N_2(g) + 3H_2O(l) \rightarrow 2NH_3(g) + 3N_2O(g)$; $\Delta H = +240$ kcal

(c) $N_2H_4(l) + H_2O(l) \rightarrow 2NH_3(g) + \frac{1}{2} O_2(g)$; $\Delta H = +36$ kcal

(d) $H_2O(l) \rightarrow H_2(g) + \frac{1}{2} O_2(g)$; $\Delta H = 68$ kcal

12. The enthalpy of combustion (in kcal/mol) of liquid N_2H_4 is -

(A) –129 (B) –138 (C) –147 (D) 147

13. The enthalpy of formation (in kcal / mol) of liquid N_2H_4 is -

(A) –11 (B) +44 (C) +11 (D) +13

14. Ammonia can be obntained according to reaction (b) or (c). Which of the following is correct regarding these reactions, for the same mass of ammonia formed.

(A) More mass of reactants is needed for reaction (c)

(B) Less mass of reactants is needed for reaction (b)

(C) More heat is needed for reaction (b)

(D) More heat is needed for reaction (c)

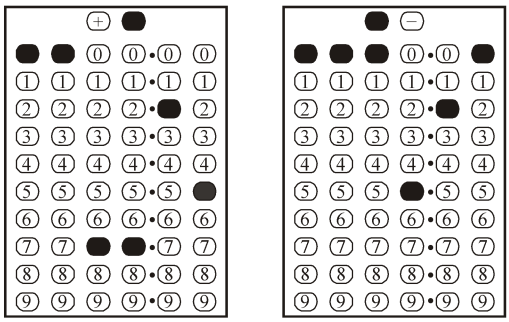
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SECTION-II: (Integer Second Decimal)

- This section contains **3** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is –77.25, 5.2 then fill the bubbles as follows.



- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

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1. Pressure over 1000 mL of a liquid is gradually increases from 1 bar to 1001 bar under adiabatic conditions. If the final volume of the liquid is 990 mL, and ΔH of the process, assuming linear variation of volume with pressure, is 'x' kJ, then the value of (x) is.

2. One mole of ideal gas undergoes following cyclic process:

(I) Isochoric heating from (P_1 V_1 T_1) to double temperature.

(II) Isobaric expansion from to double volume.

(III) Linear expansion (on PV curve) to (P_1, $8V_1$).

(IV) Isobaric compression to initial state.

If the initial temperature of the gas is 300 K and the magnitude of work done (in kcal) by the gas.

Given: $R = 2\frac{Cal}{mol-K}$

3. $2A(s) \rightleftharpoons B(g) + 2C(g) + 3D(g)$

Total pressure developed in closed container by decomposition of A at equilibrium is 12 bar at 727ºC.

If ΔG^o (kcal), of the reaction at 727ºC is (R = 2cal/mole-K, ln 2 = 0.7, ln3 = 1.1)

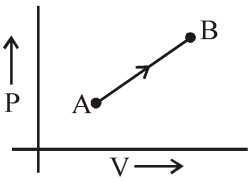
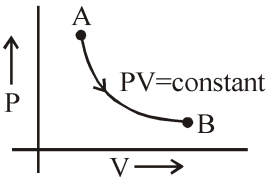
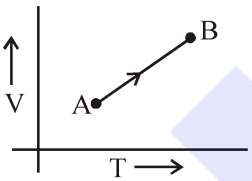
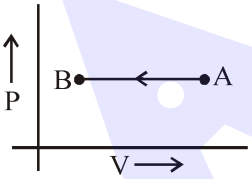
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SECTION–IV : Matrix-Match Type

This Section contains **1 question**. Question has **four graphs** (A, B, C and D) given in **Column I** and five parameters (P, Q, R, S and T) in **Column II**. Any given graph in Column I can have correct matching with **ONE** or **MORE** parameters given in Column II. For example, if for a given question, statement B matches with the parameters given in Q and R, then for the particular question, against graph B, darken the bubbles corresponding to Q and R in the ORS. **8(0)**

1. One mole of an ideal gas undergoing from state A to State-B through different processes column-I shows graph of the process & column-II shows change in parameter in these process. Match the following

Column-I		**Column-II**	
(A)		(P)	$q > 0$
(B)		(Q)	$W > 0$
(C)		(R)	$\Delta U > 0$
(D)		(S)	$\Delta H > 0$
		(T)	$W < 0$

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ANSWER KEY

SECTION–I : (i) Only One option correct Type: 3(–1)

1.	Ans.(C)	2.	Ans.(B)	3.	Ans.(A)
4.	Ans.(B)	5.	Ans.(D)	6.	Ans.(A)

(ii) One or more options correct Type : 4(–1)

7.	Ans.(A,C)	8.	Ans. (B)	9.	Ans. (B,D)
10.	Ans. (B,C,D)	11.	Ans. (A,B)		

(iii) Paragraph Type: 3(0)

12.	Ans.(C)	13.	Ans.(C)	14.	Ans.(C)

SECTION-II: (Integer Second Decimal) : 4(0)

1.	Ans. (99.50)	2.	Ans. (2.40)	3.	Ans. (17.80)

SECTION–IV : Matrix-Match Type: 8(0)

1. Ans (A) - P,R,S,T; (B) - P, T; (C) - P, R, S, T; (D) -Q

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